
Lawyers & Coders: User Manual

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1. Introduction

What the tool is:

The tool is an interactive web-based mapping platform that combines multiple environmental, health, and administrative datasets into a single interface. It allows users to visualise, filter, and analyse spatial data, and export results as PDF or Excel files for further use.

What it is used for:

The platform is designed to support environmental litigation by helping users identify patterns, generate insights, and prepare information that can be used as environmental evidence in legal and policy contexts.

Who it is for:

- Environmental lawyers and legal researchers
- Environmental NGOs and advocacy groups
- Researchers and journalists investigating agricultural or environmental issues

Datasets:

Dataset	Description	Source	Update Frequency
The Basisregistratie Gewaspercelen (BRP) link	Registers all agricultural parcels and their declared crops across the Netherlands each year. Used for EU subsidy administration and agricultural policy monitoring.	PDOK	Annual
The Basisregistraties Adressen en Gebouwen (BAG) link	The national register of all addresses and buildings in the Netherlands, including construction year, usage type, and official status.	PDOK	Continuous
Natura 2000 Areas link	EU-designated protected nature areas under the Birds and Habitats Directives. Activities within or near these	PDOK	Periodic (no fixed schedule)

	boundaries are subject to strict environmental permit requirements.		
The Natuurnetwerk Nederland (NNN) link	A national network of nature areas aimed at protecting and connecting biodiversity, defined per province under the Dutch Nature Protection Act.	Nationaal Georegister	Periodic per province
Cadastral Data link	Official cadastral parcel boundaries and sizes across the Netherlands, which is essential for identifying land ownership in legal and environmental disputes.	Nationaal Georegister	Continuous
KRD Veehouderijen link	Registered livestock farms in the Netherlands with their NH3 (ammonia), odour, and particulate matter emission values. Used in legal assessments of cumulative farm impact on nature areas.	Kernregistratie Diervverblijven	Manual export at time of download (time sensitive)
Bestrijdingsmiddelen Atlas link	Pesticide concentration measurements in Dutch surface water from the Bestrijdingsmiddelenatlas, showing which substances exceed environmental quality standards and by how much.	Bestrijdingsmiddelen Atlas	Annual
Health Facilities link	Locations of hospitals, clinics, pharmacies, and other health facilities in the Netherlands.	OSM	Continuous
School Institutions link	All registered educational institutions in the Netherlands, from primary schools (basisonderwijs) to universities.	Dienst Uitvoering Onderwijs (DUO)	Annual
Bestuurlijke Gebieden link	Administrative boundaries consist of municipal, provincial, and national borders, compiled based on the cadastral register (BRK). This dataset contains the history.	PDOK	Annual
Water Authorities Hydrography link	INSPIRE-harmonised watercourse data from Dutch water authorities (waterschappen), covering rivers, canals, and drainage channels with stream order and condition attributes.	PDOK	Periodic
WFD Surface Water link	Water Framework Directive (WFD) surface water body boundaries covering rivers, lakes, and coastal waters assessed for ecological and chemical status under EU law.	PDOK	Periodic
Waterschappen link	Administrative boundary areas of the 21 Dutch water authorities	PDOK	Periodic

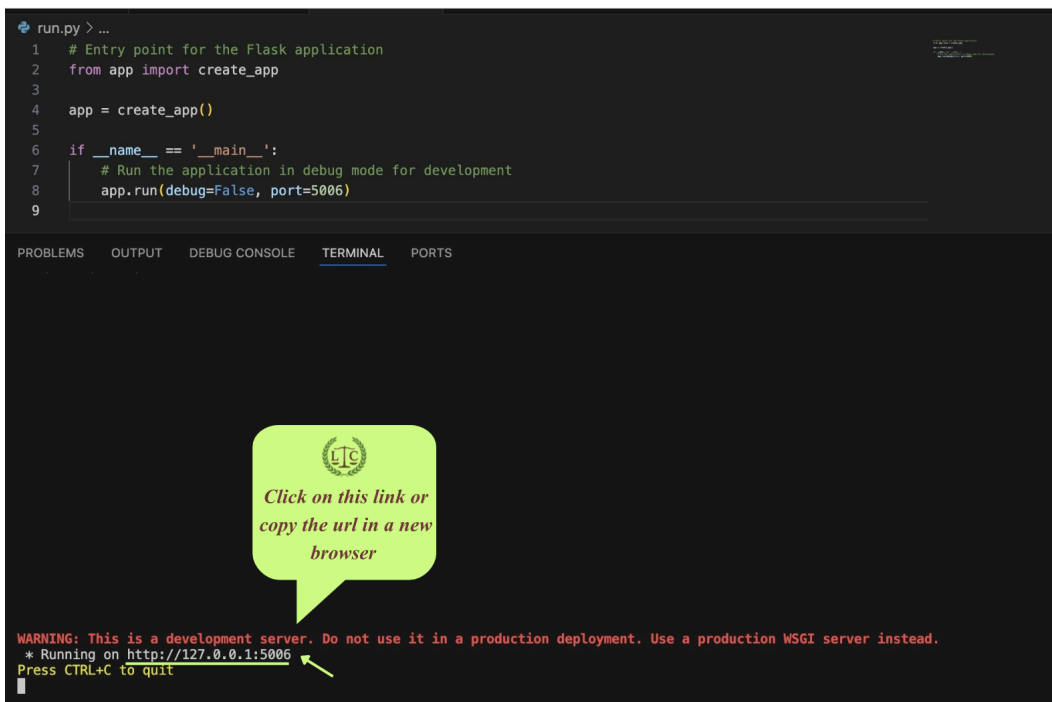
	(waterschappen), responsible for water management, flood protection, and water quality.		
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2. Getting Started

Opening the application

The application runs locally via a Python Flask server. To start it:

1. Open a terminal and navigate to the project folder
2. Run `python run.py`
3. Open a browser and go to <http://127.0.0.1:5006>



```

run.py > ...
1 # Entry point for the Flask application
2 from app import create_app
3
4 app = create_app()
5
6 if __name__ == '__main__':
7     # Run the application in debug mode for development
8     app.run(debug=False, port=5006)
9

```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

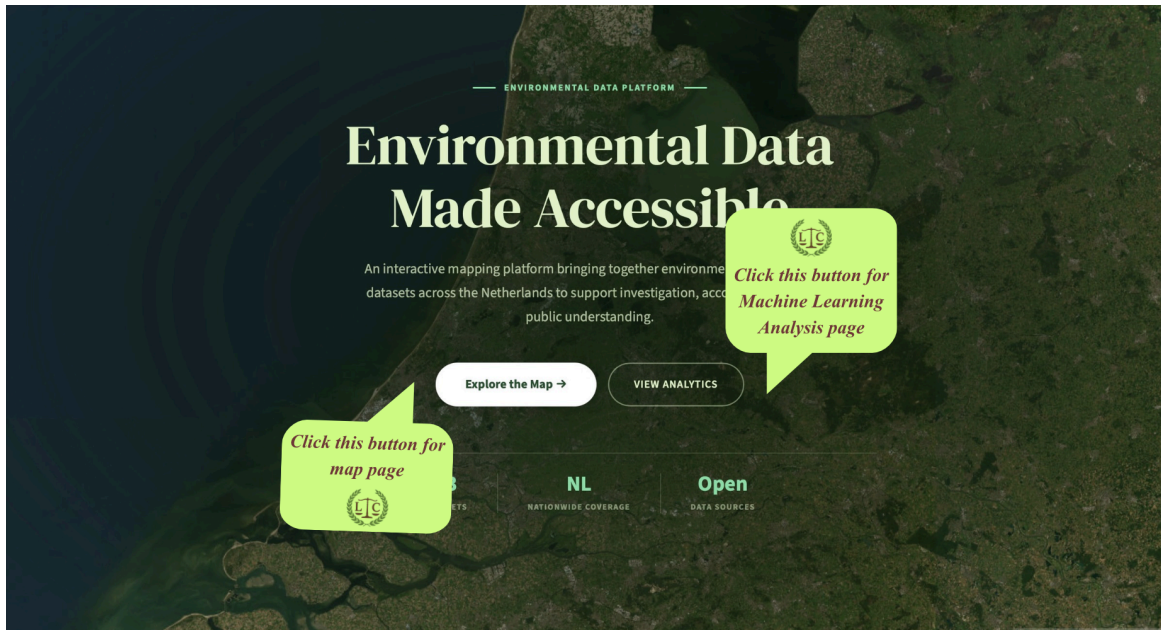
Click on this link or copy the url in a new browser

WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on <http://127.0.0.1:5006>
 Press CTRL+C to quit

Welcome: You landed on the homepage

You are faced with two buttons

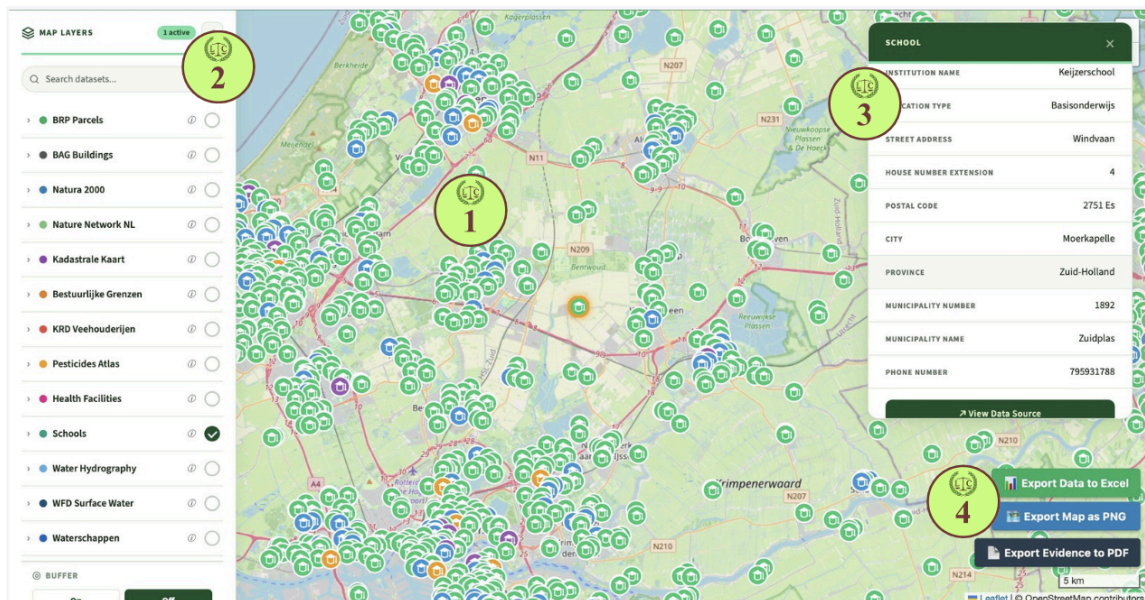
1. Explore the map
 - a. Click this if you wish to view the map
2. View Analytics
 - a. Click this if you wish to view Machine Learning analysis



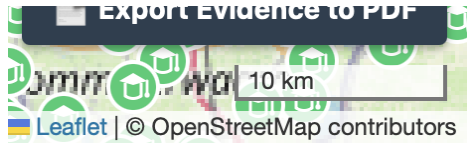
Map Interface overview

The map interface has four main areas:

1. **Main map canvas** - the interactive map, centred on the Netherlands by default
2. **Left sidebar** - the Layers Panel, where all datasets are listed to be turned on/off
3. **Info panel** - appears when you click a feature, showing its attributes and data source link
4. **Bottom-right toolbar** - export buttons (PDF, Excel/XLSX, PNG)

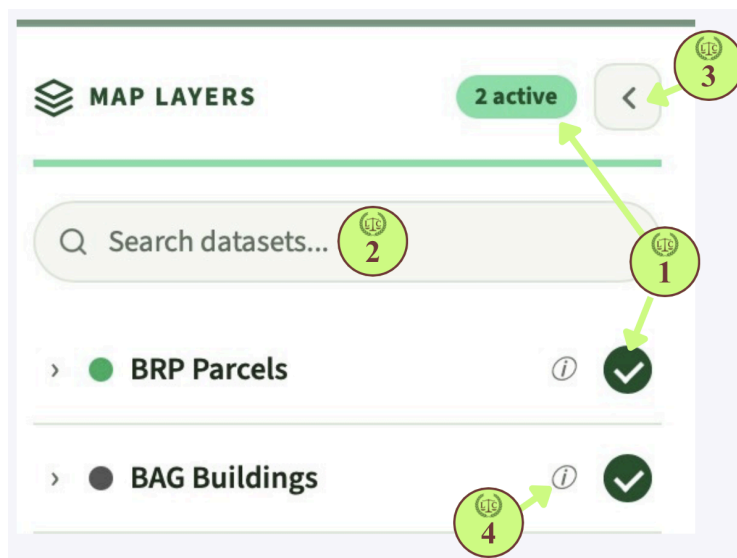


Map navigation basics

Action	How	
<i>Pan</i>	Click and drag	
<i>Zoom in/out</i>	Scroll wheel, or use the +/- buttons (top right)	
<i>Zoom scale</i>	A scale indicator (km/m) is displayed on the map at all times	
<i>Reset view</i>	Refresh the page to return to the default Netherlands view	

3. Map Layers

1. **Turning layers on/off** - Each dataset has a checkbox in the left sidebar. Tick to enable, untick to disable. A green badge at the top of the sidebar shows how many layers are currently active.
2. **Searching for a layer** - Use the **Search datasets** bar at the top of the sidebar to filter the layer list by name.
3. **Collapsing the sidebar** - Click the collapse button (<) at the top of the sidebar to hide the panel and maximise the map view. Click the tab that appears on the left edge to reopen it.
4. **Dataset information (i)** - Every layer has an (i) button next to its name. Click it to open a modal with a description of the dataset, its source, update frequency, and intended use.

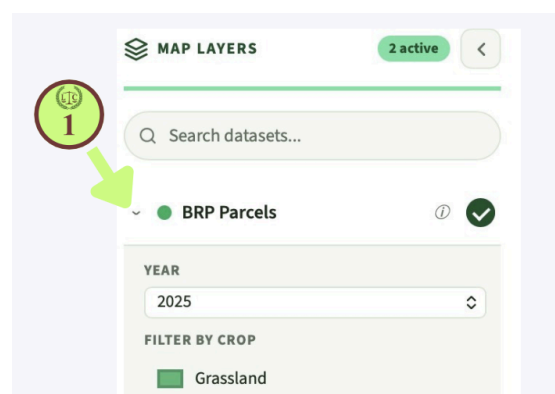


5. Layer types and what they show

Layer	What it shows	Type
BRP Parcels	Registered agricultural crop parcels by year and crop type	Polygons
BAG Buildings	All registered buildings in the Netherlands by usage type	Polygons
Natura 2000	EU-protected nature areas with configurable buffer zone	Polygons + buffer
Nature Network NL	Nature Network Netherlands protected areas with 250m buffer	Polygons + buffer
Kadastrale Kaart	Land parcel ownership boundaries	WMS tiles + click layer
Grenzen	Administrative boundaries (municipalities, provinces, national border)	Lines
KRD Livestock Farms	Registered livestock farm locations by animal type	Points
Pesticides	Pesticide concentration measurements in surface water, colour-coded by norm exceedance	Points
Health Facilities	Hospitals, clinics, and care institutions	Points
Schools	Primary and secondary schools, colour-coded by type	Points
Hydrography	Waterways and water infrastructure from water authorities	Lines/polygons
WFD Surface Water	Water Framework Directive surface water bodies	Polygons
Waterschappen	Water authority district boundaries	Polygons

6. Expanding a layer's options

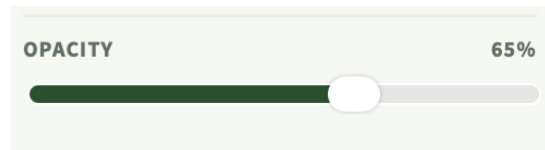
- Click the **>** arrow next to any layer name to expand its controls panel, which may include:
 - Year selector** (BRP: select data year ex: 2024)
 - Crop type filter** (BRP: toggle individual crop categories on/off)
 - Building type filter** (BAG: residential, office, industrial, etc.)



- **Animal type filter** (KRD: pigs, cattle, poultry, goats, horses, rabbits, unknown)
- **Boundary type filter** (Grenzen: municipalities, provinces, national border)
- **Buffer distance slider** (Natura 2000: adjustable from 0.1 to 20 km)
- **Municipality filter** (BRP: filter parcels by gemeente)
- **Summary table button** ("View Summary Table" - see *Section 5*)

7. *Adjusting visibility*

There is an opacity slider per layer in the current version. Slide to adjust the opacity of the selected layer.



4. **Selecting Data and Viewing Attributes**

- 1. *Clicking features*

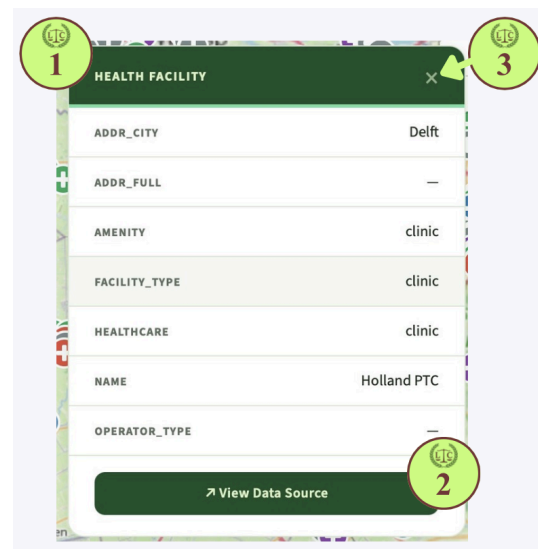
Click any visible feature on the map to open the info panel on the right side. This shows all available attributes for that feature (ex: crop type, farm name, animal count, pesticide substance, building usage).

- 2. *Data source link*

At the bottom of every info panel is a "View data source" button linking directly to the official source (PDOK, KRD, HOTOSM, Bestrijdingsmiddelenatlas, DUO, etc.). → this link is important for legal evidence; it shows where the data originates.

- 3. *Closing the info panel*

Click the **X** button on the panel, or click any empty area on the map.



- *Creating a buffer around a feature*

When you click a feature, a "Create buffer" option appears. Select a radius in kilometres to draw a buffer zone around that feature. This is useful for proximity analysis (ex: "is this farm within 500m of a Natura 2000 area?").

5. Analysis Tools

Buffer zones

Buffers can be created:

1. **Click a feature** → **Create buffer** - draws a circular buffer around a point or polygon feature at a chosen radius

Once a buffer exists, all other active layers are spatially filtered to show only features that intersect the buffer zone.

Summary tables (pivot tables)

Every layer has a "*View Summary Table*" button in its expand panel. This opens a popup showing aggregated statistics for the features currently loaded in the viewport. For example:

- BRP: total hectares per crop type in the current view
- KRD: farm count by animal type
- BAG: building count by usage type
- Pesticides: substances exceeding environmental norms

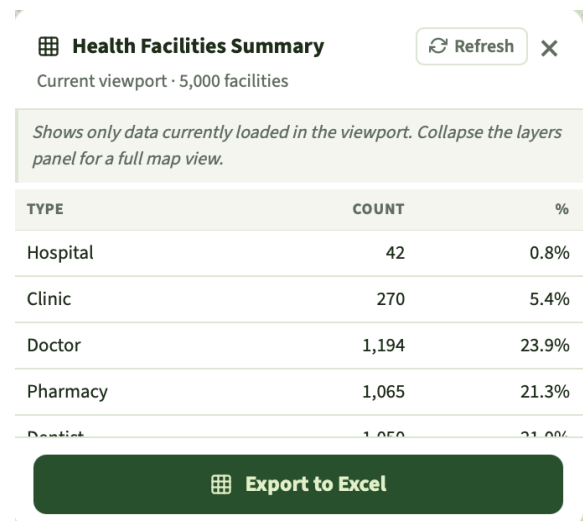
Summary tables update when you pan or zoom the map, and a Refresh button forces a manual update.

BRP crop trend analysis

The BRP layer has an additional **"Trend"** button that shows year-by-year changes in crop type distribution. This can be useful for demonstrating long-term land use shifts.

BAG buffer summary

When a buffer is active and the BAG Buildings layer is enabled, the summary automatically shows how many residential buildings fall within the buffer zone.



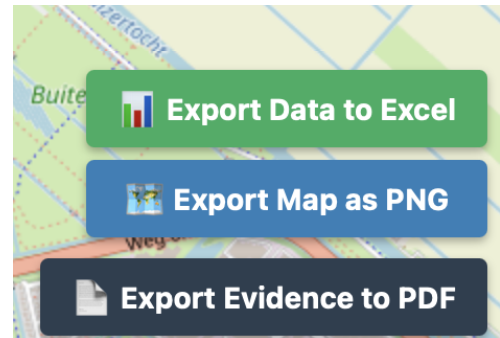
TYPE	COUNT	%
Hospital	42	0.8%
Clinic	270	5.4%
Doctor	1,194	23.9%
Pharmacy	1,065	21.3%
Dentist	1,050	21.0%

6. Export Functions

Export data to Excel (XLSX)

Click the “**Export Data to Excel**” button. This sends the currently active and visible data to the server, which returns a structured .xlsx file. The file contains:

- One sheet per active dataset, with all attributes for features in the current view (or buffer zone if one is active)
- Each sheet includes a data date row and an export date row at the top
- A dedicated Data Sources sheet listing the official source URL and publication date for every included dataset



Export map as PNG

Click the “**Export Map as PNG**” button. This saves a screenshot of the current map view.

Export map (PDF/image)

Click the “**Export Evidence to PDF**” button (bottom right). The current map view is captured as an image and embedded into an A4 landscape PDF. The PDF includes:

- The map image
- Attribution text listing the active data sources
- The export date

7. Example Use Cases

Agricultural impact on a Natura 2000 area

1. Enable Natura 2000 and set the buffer
2. Enable KRD Livestock Farms and use the Natura 2000 distance filter to show only farms within that zone
3. Click individual farms to view animal type and farm registration details
4. Open the KRD Summary Table to see total farm counts by animal type within the buffer
5. Export to Excel → the Data Sources sheet will document the KRD and PDOK Natura 2000 sources with dates

Pesticide contamination near residential areas

1. Enable WFD Surface Water or Hydrography to locate the water body of interest

2. Enable Pesticides (features are colour-coded by norm exceedance severity)
3. Click a measurement point to see the substance name, concentration, and norm threshold
4. Enable BAG Buildings filtered to Residential and draw a buffer around the water body
5. The BAG buffer summary will show the count of residential buildings within the affected zone
6. Export to PDF and Excel

Land use change over time (BRP)

1. Enable BRP Parcels and select a year from the dropdown
2. Use the crop type filters to focus on specific categories (e.g. show only Grassland and Maize)
3. Use the Municipality filter to narrow to a specific gemeente
4. Open the BRP Trend tool to see how crop composition has shifted year-over-year
5. Export the summary table to Excel for use in a report

Proximity of schools or health facilities to emissions sources

1. Enable KRD Livestock Farms and Schools or Health Facilities
2. Click a farm and create a 1km buffer
3. The map will filter to show only schools/health facilities within that buffer
4. Click each facility to view its name and type in the info panel
5. Export to Excel → the schools sheet will list all affected facilities with coordinates

8. Troubleshooting

Layers don't load or appear blank

- *Check your internet connection* - several layers (Natura 2000, NNN, Hydrography) are fetched live from PDOK's servers
- *Zoom in* - BRP and Kadastrale Kaart only load data within the current viewport; zoom in to a specific area first
- *Refresh the page* - some layers have a fallback to the local database if the live API fails; a page refresh triggers this fallback
- *PDOK outages* - if multiple layers fail simultaneously, PDOK's servers may be temporarily unavailable. Check pdok.nl for service status.

The map is slow or freezing

- *Disable unused layers* - having many polygon-heavy layers active simultaneously (BRP + BAG + Kadastral) is the most common cause of lag
- *Zoom in further* - the app loads data only for the current viewport; a zoomed-out view loads far more features

- Use WMS layers - Natura 2000, NNN, and Kadastrale Kaart use pre-rendered tile images when active; these are faster than raw data layers

The Excel export is empty

- Make sure at least one data layer is enabled before clicking export
- If a buffer is active, only data intersecting the buffer is exported. Check that the buffer actually overlaps your data

Feature clicks don't open the info panel

- Natura 2000 and NNN use WMS tiles for visual rendering. Click interaction is handled by a separate invisible GeoJSON layer that loads on zoom. Zoom in and wait for the data layer to load before clicking

9. Known Data Limitations

1. KRD farm locations are approximate

Farm coordinates in the KRD dataset are stored in the Dutch RD New coordinate system (EPSG:28992) and reprojected to WGS84 for display. The coordinates represent the registered address location of the farm, not necessarily the precise location of animal housing units. For precise emission source locations, individual stallen (housing unit) data should be consulted.

2. KRD coverage is limited to three provinces

The KRD dataset loaded in this tool covers only Gelderland/Twente, Limburg, and Noord-Brabant. Farms in other provinces are not included.

3. KRD data reflects conditions at time of export

The KRD is not available via a live API. The data reflects the state of farm registrations at the moment the CSV files were manually downloaded from krd.igoview.nl. Farms that have opened, closed, or changed animal type since that date will not be reflected.

4. School coordinates are geocoded, not surveyed

School locations were derived by geocoding school addresses from DUO open data, while the coordinates were derived from PDOK's street address to coordinates converter API. Geocoding converts a street address to an approximate coordinate using a mapping database. Schools on corner plots, schools with ambiguous addresses, or schools that have recently moved may appear slightly offset from their true location.

5. Healthcare facility CRS is assumed

The health facilities dataset (HOTOSM) does not always include a declared coordinate reference system. Where no CRS is declared, the tool assumes WGS84 (EPSG:4326), which is standard for OpenStreetMap exports. In the rare case this

assumption is incorrect, facilities could appear in the wrong location.

6. *Healthcare and school data are not authoritative government sources*

Health facilities data comes from OpenStreetMap (HOTOSM), which is community-edited and not maintained by a government authority. Schools data comes from DUO, which is authoritative, but coordinates are derived rather than registered. Neither should be used as the sole basis for a legal claim about a specific facility's location without cross-referencing an official source.

7. *BAG and Cadastral data reflect the database snapshot date*

Although both Kadaster datasets are continuously updated at source, the BAG Buildings data in the local database only reflects conditions as of the last ETL reload. The Kadastrale Kaart layer, however, uses live WMS tiles and always shows current Kadaster data.

8. *Pesticide measurements are point samples, not continuous coverage*

The Bestrijdingsmiddelenatlas data represents discrete measurement points in surface water. The absence of a measurement point near a location does not mean contamination is absent. It may simply mean that location has not been sampled.

9. *Buffer zones are geometric approximations*

Buffer distances are calculated as straight-line distances on the map plane, not as travel distances or terrain-adjusted distances.

10. *Viewport-based loading*

BRP, BAG, Kadastrale Kaart, and KRD data load only within the current map viewport. Features just outside the visible area are not included in summary tables or Excel exports. Always zoom to fully encompass your area of interest before exporting.

10. System Requirements

For end users (running a hosted version)

- A modern web browser: Google Chrome (recommended), Firefox, Edge, or Safari
- JavaScript must be enabled
- A stable internet connection is required for layers that stream live data

For self-hosting / developers

The application requires the following to be installed and configured:

Requirement	Version / Notes
Python	3.11 or 3.12 recommended
PostgreSQL	13 or later, with the PostGIS extension enabled

Environment file	A .env file in the project root containing DATABASE_URL=postgresql://user:password@host:port/dbname
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First-time database setup

1. Create a PostgreSQL database and enable the PostGIS extension: CREATE EXTENSION postgis;
2. Create and populate the .env file with your database connection string
3. Run the ETL scripts for each dataset
4. Start the application with python [run.py](#)

11. Legal Disclaimer

This tool presents data sourced from Dutch government open data platforms and third-party contributors. The following points must be understood before using any output from this tool in legal or court proceedings:

1. Data accuracy is not guaranteed. While all datasets are sourced from official Dutch government platforms or recognised open data providers, we do not verify the accuracy or completeness of the underlying data. Errors in source data will be reflected in this tool.
2. Datasets are loaded at a point in time and are not automatically refreshed. The data you are viewing may not reflect current conditions.
3. Buffer zones are not legally defined distances. The buffer tools in this application are for spatial analysis and visualisation purposes only. They are geometric approximations and do not constitute legally defined protection zones, permit boundaries, or regulatory thresholds.
4. This tool is not a substitute for professional advice; it is designed to assist in identifying and documenting spatial relationships. It does not constitute legal advice, environmental assessment, or regulatory guidance. Outputs should be reviewed by a qualified environmental lawyer, or specialist before being submitted as evidence.
5. The export date is not the data date. The date in the filename of exported PDFs and Excel files reflects the date the export was generated, not the date the underlying data was collected or published.